

Building the operational flow as an agentic AI system

This plan takes the nine-stage business process and the ten evaluation scenarios already documented and maps each one to a concrete technical implementation — which component performs it, whether that component is a deterministic lookup or an AI agent, and the exact decision logic that routes a case to one of the ten outcomes. It's the direct technical counterpart to the operational walkthrough and scenario reference.

Solution approach

The business process reads as a single linear flow, but the correct technical implementation is **not one large AI agent making every decision**. Most of the nine stages are either pure data-handling (no AI needed) or a deterministic rule check (a yes/no lookup against a database). Only three points in the process genuinely need an AI agent's judgment: reading unstructured documents, writing a summary, and reasoning about whether free-text evidence satisfies a clinical requirement. Everything else — including most of what decides the outcome in the ten scenarios — is ordinary, auditable business logic.

This matters operationally: a rule-based check is fast, free, 100% consistent, and easy to audit. An AI agent's reasoning is powerful but should be reserved for the steps that genuinely require reading and interpreting unstructured text. The plan below keeps that split explicit throughout, using three labels:

DETERMINISTIC	A rule, a database lookup, or a calculation — same input always gives the same output, no AI model involved.
AGENTIC (AI)	A step that requires an AI model to read, interpret, or reason about unstructured text.
HUMAN	A step performed by appeals staff, not automated at all.

Part 1 — Stage-by-stage technical mapping

Each row corresponds directly to a stage in the operational walkthrough.

Stage	Business activity	Type	Technical component
1. Appeal received	Case opened from an email or upload	DETERMINISTIC	Email connector / upload endpoint — captures and time-stamps, no interpretation.
2. Documents organized	All text pulled into one working file	DETERMINISTIC	Document text extraction service (OCR for scanned files, direct read for typed ones).
3. Privacy protected	Unneeded personal details removed	DETERMINISTIC	PII/PHI detection & masking service, runs before any AI model sees the text.
4. Key details identified	Facts extracted from free text; identity confirmed	AGENTIC (AI)	Intake Agent — reads unstructured text, produces structured fields, resolves identity when a member ID is missing.
5. Matched to records on file	Denial, benefit, and history records located	DETERMINISTIC	Three keyed lookups (exact match by member ID / denial reference) — no AI needed, this is a database read.
6. Summary prepared	Case condensed into a short summary	AGENTIC (AI)	Summarization Agent — reads the full case and writes a plain-language summary.
7. Evaluated against requirements	Case checked against policy; outcome decided	AGENTIC (AI)	Evaluation Agent — combines the deterministic checks in Part 2 with AI-based reading of the applicable policy for the clinical judgment.
8. Reviewed by staff	A person confirms or adjusts the recommendation	HUMAN	Reviewer dashboard — surfaces everything gathered; records the human decision.
9. Closed and recorded	Final decision and history saved	DETERMINISTIC	Case finalization — writes the permanent record, no AI involved.

Of nine stages, three are AI agents, five are deterministic logic or plumbing, and one is a human decision. That ratio is intentional — it's what keeps the system fast, explainable, and cheap to run at volume.

Part 2 — The decision logic plan

This is the logic inside Stage 7 that decides which of the ten scenarios a case lands in. It runs as an ordered sequence of checks — cheapest and most certain first, the AI-based clinical judgment last — and stops at the first check that produces a conclusive answer. This ordering matters: it means a case that fails on something simple (a missed deadline, a lapsed eligibility date) never wastes an AI reasoning call on a clinical question that was never going to matter.

check 1 — deterministic

Can the member be confidently identified?

No → Cannot process (Scenario 8) — hold, request member ID or date of birth. Stop.

↓ yes, continue

check 2 — deterministic

Does this case number already have an open, unresolved appeal?

Yes → Duplicate (Scenario 9) — merge into the existing case. Stop.

↓ no, continue

check 3 — deterministic

Has this denial already been through one internal appeal that was upheld?

Yes → Escalate (Scenario 10) — route to external review. Stop.

↓ no, continue

check 4 — deterministic

Was the appeal filed within the plan's appeal window?

No, and no good-cause reason given → Deny — filed after deadline (Scenario 6). Stop.

↓ yes, continue

check 5 — deterministic

Was the member's coverage active on the date of service?

No → Deny — not eligible on date of service (Scenario 5). Stop.

↓ yes, continue

check 6 — deterministic

Is the requested service a covered benefit under this plan?

No, and the documented exception condition isn't met → Deny — not a covered benefit (Scenario 3). Stop.

↓ yes, continue

check 7 — deterministic

Does the request stay within the member's remaining annual benefit allowance?

No, and none of the required override evidence is present → Deny — limit exceeded, no override (Scenario 4). Stop.

No, but some (not all) override evidence is present → Needs more information (Scenario 7) — name the specific missing item. Stop.

↓ yes / fully justified, continue

check 8 — agentic (AI)

Does the case meet every criterion in the applicable clinical policy?

Thresholds clearly not met, no new evidence in the appeal → Deny — clinical requirements not met (Scenario 2). Stop.

Most criteria met; one specific, identifiable item is missing → Needs more information (Scenario 7). Stop.

All criteria met, or the only gap is simply undocumented (not contradicted) → Approve (Scenario 1), flag any undocumented item for the reviewer.

Why the AI step comes last: check 8 is the only step that requires genuine judgment — reading a policy bulletin and deciding whether free-text clinical evidence satisfies it. Every check before it is a simple lookup a rules engine can answer with certainty, so routing those first means the AI model is only ever invoked on cases that have already cleared every procedural and eligibility hurdle — fewer calls, faster answers, and a clear audit trail showing exactly which rule (not "the AI") produced any denial before check 8.

Part 3 — Agent design

Three AI agents, each scoped to exactly one of the judgment-requiring stages. All three run inside one orchestrated workflow, sharing the case record as shared state, so each can be built and tested independently before being wired together.

STAGE 4 • CHECK 1

Intake Agent

Reads the combined, privacy-protected document text and produces the structured case fields. Also performs identity resolution when a member ID is missing but a name and date of birth are present — this is what separates Scenario 8 (cannot process) from a normal extraction: the agent must actively decide whether it has *enough* to proceed, not just extract whatever is there.

READS
document text (privacy-protected)

CALLS
member lookup tool (for name+DOB resolution)

PRODUCES
member, provider, request, diagnosis fields

GUARDRAIL
never fabricates a missing field — reports it as missing instead

STAGE 6	
Summarization Agent	
Reads the full case — structured fields plus every source document — and writes the short plain-language summary a reviewer reads first. Purely descriptive; makes no determination about the outcome.	
READS	PRODUCES
structured case + full document text	a short reviewer-facing summary

STAGE 7 • CHECK 8 ONLY	
Evaluation Agent	
Owns just the last step of the decision logic plan — checks 1 through 7 are handled by a plain rules engine before this agent is ever invoked. Given the applicable policy and the case's clinical evidence, it retrieves the relevant policy passage, checks each numbered criterion, and produces a recommendation with a citation for every claim it makes.	
READS	PRODUCES
case summary + structured case + retrieved policy text	decision, reasoning, per-criterion findings, citations
CALLS	GUARDRAIL
policy retrieval tool (semantic search over policy bulletins)	every claim must cite a retrieved passage — no citation, no claim

Part 4 — Implementation phases

Each phase produces something demoable and is validated against specific scenarios from the reference document before moving on.

Phase 1 rules engine

Build checks 1-7 first, with no AI at all

These are ordinary business rules against a database — they can be built, tested, and demoed before a single AI agent exists. Validates Scenarios 3, 4, 5, 6, 8, 9, 10 entirely on their own.

Phase 2 Intake Agent

Add document reading and field extraction

Wire the Intake Agent in front of the Phase 1 rules engine. Validates that check 1 (identity) correctly triggers Scenario 8 from real unstructured input, not just test data with the field already populated.

Phase 3 Evaluation Agent

Add the clinical judgment step

Build check 8 against a small starting set of policy bulletins. This is the only phase that needs a policy library in place first. Validates Scenarios 1, 2, and 7.

Phase 4 Summarization Agent reviewer view

Add the summary and the human review step

Lowest-risk phase — the summary doesn't affect the decision logic at all, it just makes the case readable for the person confirming it.

Phase 5 full regression

Run all ten scenarios end to end

Every scenario in the reference document should be re-run through the complete chain before this is considered done — including the negative ones, since a system that only gets tested on approvals will quietly drift toward approving everything.

Part 5 — What still needs to be built for full test coverage

The intake test data and reference files built earlier only cover Scenarios 1, 7, and 8 with real example files. To fully validate checks 2 through 6 (duplicate, escalation, deadline, eligibility, coverage) and the harder version of check 7 (Scenario 4), matching example records need to be added for:

- Robert Yeung, Priya Anand, David Okafor, Helena Brooks, and Thomas Reyes — the five denial examples from the scenario reference — as denial, eligibility, and (where relevant) utilization records in the same format as the existing reference data.
- A duplicate-submission test pair (two near-identical emails referencing the same case number).
- An already-upheld appeal record for the escalation check (Angela Ruiz), including the date and outcome of the first internal review.

This is a natural next deliverable — the same structure as the reference data already built, just extended to cover the five scenarios that don't yet have backing test files.

This plan is the technical counterpart to the operational walkthrough and the scenario reference — the nine stages and ten outcomes are identical to those documents; what's new here is which component performs each one and the exact order in which the decision logic runs.